

STA556: Statistics and Data Science Computing Workflows

Fall 2026

TTh 2:20 – 3:35 pm

Prerequisites

Admission to MS Statistics and Data Science

Catalog description

This course equips graduate students with the advanced computational tools and programming skills essential for modern statistics and data science. Students develop proficiency in writing robust, efficient code using industry-standard languages, including (but not limited to) Python, R, and Julia. The curriculum focuses on the practical implementation of statistical methods, covering complex data structures, data wrangling from varied sources (such as SQL and APIs), and professional-quality data visualization. Students will master algorithmic efficiency by implementing statistical algorithms—including Monte Carlo simulations, bootstrapping, and numerical optimization—from scratch. Emphasis is placed on constructing reproducible research workflows and automated reports using literate programming tools like Quarto, R Markdown, or Jupyter. Additionally, the course covers code debugging and profiling to optimize time and memory usage, alongside critical discussions on data ethics and algorithmic bias.

Course Content

1. **Course Introduction & Environment Setup**
 - Command line basics (Bash/Zsh)
 - Git and GitHub for version control
 - IDE setup (VS Code, Jupyter, RStudio)
2. **Advanced Data Types & Structures**
 - Mutable vs. immutable objects
 - Memory models
 - Static vs. dynamic typing
 - List and dictionary comprehensions
3. **Data Transformation & Wrangling**
 - DataFrames (Pandas/DataFrames.jl)
 - Indexing, slicing, and filtering
 - Joining, merging, and concatenating data
 - Reshaping data (pivoting/melting)
 - Missing-data strategies
4. **Input/Output & External Data Sources**
 - SQL databases
 - Web APIs
 - JSON and HTML data
 - Web scraping
 - Binary file formats (Parquet/HDF5)
5. **Exploratory Data Analysis & Visualization**

- Grammar of Graphics
 - Visualizing continuous and categorical data
 - Customizing visual aesthetics
 - Publication-quality visualization
6. **Modular Programming & Functions**
 - Pure functions
 - Variable scope
 - Functional programming
 - Map/reduce/filter operations
 7. **Reproducibility & Reporting**
 - Literate programming
 - Quarto
 - Jupyter
 - R Markdown
 - Automated and dynamic reports
 8. **Matrix Computation & Linear Algebra**
 - Broadcasting
 - Vectorized operations
 - Matrix decompositions
 - SVD and QR decomposition
 - Linear algebra for statistical computing
 9. **Simulation Methods**
 - Pseudo-random number generation
 - Inverse transform sampling
 - Monte Carlo methods
 10. **Resampling Methods**
 - Bootstrapping
 - Permutation tests
 - Implementing resampling methods from scratch
 11. **Numerical Optimization**
 - Root finding
 - Newton-Raphson method
 - Gradient descent
 - Loss-function minimization
 - Maximum likelihood estimation (MLE)
 - Ordinary least squares (OLS)
 12. **Debugging & Testing**
 - Unit testing (pytest/Test.jl)
 - Defensive programming
 - Exception handling
 13. **Profiling & Computational Efficiency**
 - Time profiling
 - Memory profiling
 - Identifying and optimizing bottlenecks
 - Parallel computing
 - Compilation and optimization (Numba/Julia types)
 14. **Data Ethics**
 - Algorithmic bias
 - Fairness

- Accountability
- Transparency in statistical modeling

Course Structure

The class will utilize a hybrid format of lectures and project-based learning. Students will be required to both conceptually understand data science concepts as well as be able to implement them using their own written code and computer hardware.

Reading from various sources will be assigned to be completed at home prior to the topic being covered in class.

Course Learning Outcomes

Upon successful completion of the course, the student will be able to:

1. Manipulate and clean complex datasets from various sources (e.g., CSV, SQL, APIs) into formats suitable for analysis.
2. Develop custom functions and scripts to automate statistical tasks and ensure modularity.
3. Implement statistical algorithms, such as Monte Carlo simulations, bootstrapping, and optimization techniques, using matrix-based programming.
4. Create professional-quality data visualizations to explore patterns and communicate findings.
5. Debug and profile code to identify errors and optimize computational efficiency (time and memory usage).
6. Construct reproducible research workflows and reports using literate programming tools (e.g., R Markdown, Quarto, or Jupyter).
7. Configure and manage professional development environments using version control (e.g., Git/GitHub) and Integrated Development Environments (IDEs) to facilitate collaborative coding.
8. Execute basic system operations via the command line to automate file management and script execution.
9. Implement idiomatic programming patterns, such as comprehensions and type-hinting, to write readable and performant code.

Attendance

Attendance is expected for all class sessions. Missing more than four (4) class periods in a semester can constitute the loss of a letter grade. No provisions will be made for missed work except for absences deemed institutionally acceptable.

Assignments/Assessments

The following items will form the assessment for this course:

- 2 x mid-term exams (2 x 15% each)
- 4 x Homework Assignments (40% total)
- Final Exam (25%)

Grading Policy

The grade ranges will be:

A	100-90%
B	89-80%
C	79-70%
D	69-60%
F	<60%

The instructor reserves the right to change grade ranges, nominally to lower the cutoffs for each grade range.

Technology

This course requires use of the python, R, and/or Julia programming languages. These are available free of charge and are pre-installed on many operating systems. They can be accessed in the computer lab AMB 222 and on many/most other computers found around campus.

Changes to the Syllabus

The instructor reserves the right to make changes in the syllabus as needed, including changes in exam dates. Such changes will be announced in class and posted to the course website.

Textbook and Course Materials

The primary references for this course are:

1. **Python Data Science Handbook** by Jake VanderPlas.
 - The core data science workflow. It covers **NumPy** (matrix operations), **Pandas** (data wrangling), **Matplotlib** (visualization), and **Scikit-Learn** (machine learning).
 - *Format:* Available for free online (<https://jakevdp.github.io/PythonDataScienceHandbook/>) or in print.
2. **Python for Data Analysis (3rd Ed)** by Wes McKinney.
 - Data manipulation in python. A suggested reference for cleaning, processing, and restructuring complex datasets.
 - *Format:* Open access online (<https://wesmckinney.com/book/>) or in print.
3. **R for Data Science (2nd Ed)** by Wickham, Çetinkaya-Rundel, and Grolemund.
 - Data wrangling, visualization, and reproducible reporting.
 - *Format:* Available for free online (<https://r4ds.hadley.nz/>) or in print.
4. **Julia Data Science** by Jose Storopoli, Rik Huijzer, and Lazaro Alonso.
 - The core data science workflow. It covers **DataFrames.jl** (wrangling), **Makie.jl** (visualization), and **Flux.jl** (machine learning). It is modern, open-source, and specifically designed to be an accessible entry point.
 - *Format:* Available for free online (<https://juliadatascience.io/>) or in print.

Supplementary materials will be provided on canvas including journal articles, academic conference presentations, links to video explanations and additional lectures, and papers from additional textbooks.

Career Readiness Skills

In every class you take at NAU, you learn professional skills that can support your future career. There are several ways that this course can help you meet and excel at your job goals and life desires. Below is a list of in-demand skills from National Association of Colleges and Employers (NACE) you will practice in this class:

1. Communication: Demonstrate the ability to articulate mathematical concepts clearly and concisely, whether through written explanations, oral presentations, or visual representations, ensuring comprehension by peers.
2. Critical Thinking: Demonstrate the ability to solve mathematical problems by considering the context in which they arise, ensuring that solutions are relevant and applicable to real-world situations.
3. Professionalism: Uphold academic integrity and accountability in mathematical assignments, demonstrating honesty and ethical behavior in the completion of individual and group tasks.

4. Teamwork: Collaborate actively with classmates to achieve common mathematical goals, working collectively on assignments, projects, or problem-solving exercises to enhance the overall learning experience.

Some Career Readiness resources:

- LinkedIn CEFNS Career Development www.linkedin.com/in/cefns-career-development-072715233
- LinkedIn NAU Career Development <https://www.linkedin.com/company/nau-career-development/>
- Handshake: <https://nau.joinhandshake.com/login>
- Udemy: Online courses and career searching advice <https://in.nau.edu/its/udemy/>
- Log in with your NAU email account and search 'NAU Career Steps'
- O*net Online: Occupation exploration reports <https://www.onetonline.org/>
- Labor Market Insights <https://frankecareer.nau.edu/labor-market-insights/>

NORTHERN ARIZONA UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

Fall 2026 Syllabus Policies

Course Prerequisites and Placement

Prior to enrollment in a course in the Department, a student must have completed the course prerequisites or have proper placement for the course. It is the student's responsibility to check that they are properly enrolled in a course and to drop the course if they are not. Failure to do so could result in the student receiving no credit for the course. The department may cancel a student's registration in a course in which they are not properly enrolled. It is students' responsibility to monitor their own enrollment.

Administrative Drops

A student may be administratively dropped through the 7th business day (**September 1, 2026**) from a course by the instructor if the student meets either of the two following criteria: (1) has not met the prerequisites for the course as stated in the academic catalog, or (2) is absent one or more times from class during the first four business days of the semester.

Class Attendance

Students are expected to assume full responsibility for class attendance and are accountable for work missed because of absences. Instructors are under no obligation to make special arrangements for students who have been absent unless such absence has been excused by a formal institutional excuse. Institutional excuses permit a student to be absent from classes to represent the University in athletics and extracurricular or academic activities. Institutional excuses must be hand-delivered to the instructor and arrangements made for the work missed prior to the planned absence from class.

Adding/Dropping/Auditing a Course

The last day to add a class is **September 2, 2026**. The last day to add to a *waitlist* is **September 1, 2026**. The last day to drop/delete a course (*without the class appearing on your transcript*) is **September 2, 2026**. The last day to drop a course (and receive a **W**) is **December 4, 2026**. Academic policy requires that a student who never attended class or stopped attending class receive an **F** should the student fail to officially drop the course. The deadline to change from credit to audit or vice versa is **September 2, 2026**. Once a student has registered and completed a class as an auditor, the audit grade cannot be changed to a credit-earning grade. The grade of **AU** is awarded to auditors for satisfactory attendance. More information can be found in the [NAU Catalog](#).

The Grade of Incomplete

A student, who, for reasons beyond the student's control, is unable to complete course requirements during the instructional period, may make a request to the instructor for a grade of Incomplete (I). The student must be passing the course and must have completed a majority of the course. Before a grade of I can be given the student and instructor must complete the official department form indicating the work to be completed, as well as the date(s) by which the work must be completed. All work must be completed within one year. After one year, a grade of I automatically reverts to a grade of F.

Final Examinations

Final examinations are required in all classes and must be given at the scheduled times and dates indicated in the university final exam schedule. An exception to the official final examination schedule can be made if a student is scheduled to take more than two examinations in a 24-hour period. For more information, see the [Final Exam Schedule](#) (MS Word file).

NAU Policy Statements

Additional institutional Syllabus Policy Statements can be found on the [Academic and Faculty Affairs Syllabus Requirements page](#).

Department Policy on Use of Portable Electronic Devices

Cell phones, mp3 players and portable electronic communication devices, including but not limited to smart phones, cameras, and recording devices must be turned off and inaccessible during in-class knowledge assessments. Any violation of this policy will be treated as a violation of the student academic integrity policy.